Canada Post Small-Business Initiative

Group Members –

Charanjeet Singh
Roshan Reji
Mayur Utekar
Siddhant Berry
Hariram Manickavasagam

TABLE OF CONTENTS

INTRODUCTION

CANADA POST
POSTES CANADA

Key Challenge:

Large and diverse potential customer pool for postage meters within the SMB sector.

Project Focus:

Leverage data analysis and insights for highly targeted marketing campaigns.

OBJECTIVE

- Analyze Historical Data: Thoroughly examine Canada Post's previous campaign data to uncover patterns and correlations that distinguish successful postage meter adopters within the SMB sector.

- Build a Predictive Model: Develop a robust predictive model that leverages insights from the analysis to accurately identify SMBs with the highest potential for postage meter adoption.

- Refine Targeting Strategy: Use the predictive model to create a targeted marketing strategy that prioritizes high-potential SMB leads for Canada Post's future postage meter campaigns.

DATA OVERVIEW

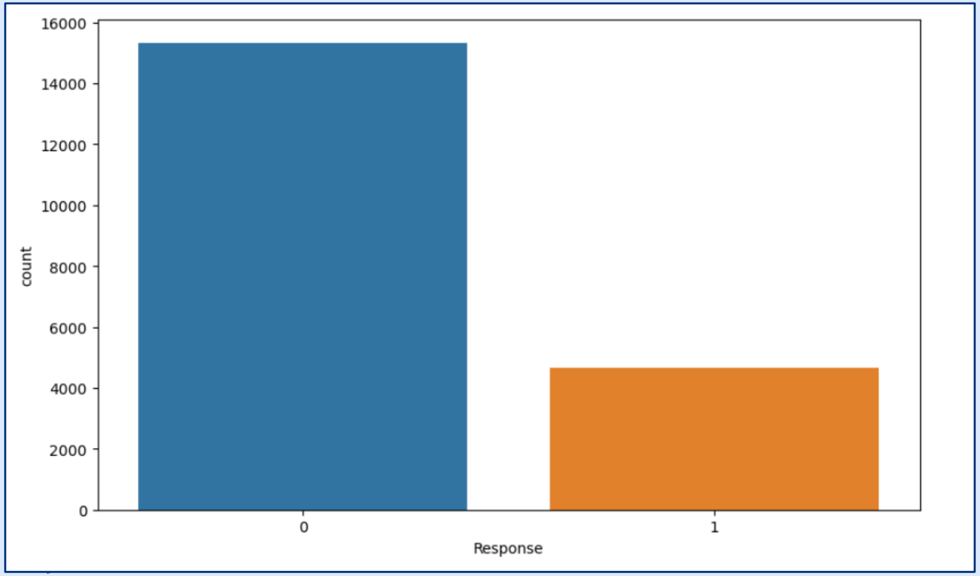
Data Sources:

Respondent Dataset: Contains information on SMBs who responded positively to a previous postage meter marketing campaign.

Non-Respondent Dataset: Contains information on SMBs who did not respond to a previous postage meter marketing campaign.

Data Preparation:

Datasets were combined to enable comparative analysis.
A "Response" column was created:
Responders coded as "1“
Non-responders coded as "0"



DATA PREPARATION

Key Variables:

- Quantitative: Sales, Employees_Here, Employee_Total
- Categorical: City, Province, Four_Digit_SIC, SMSA

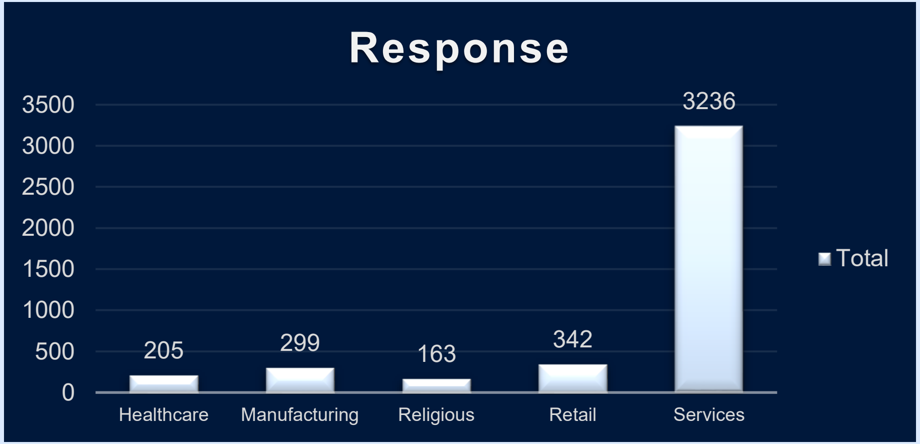
Feature Engineering

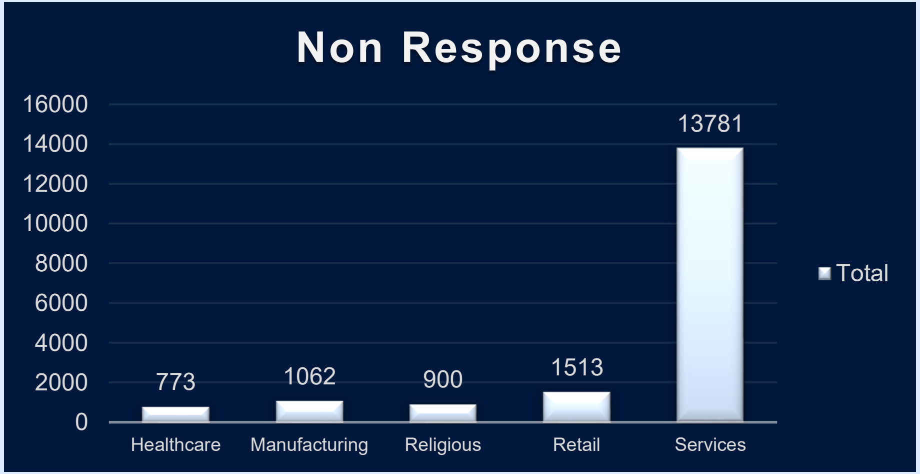
- Ratio Calculations: Enhanced analysis by creating ratios (e.g., Sales_Ratio_City) to normalize categorical and ordinal data.
- Keyword Extraction: Derived a "Business_Category" variable with 20 categories, providing insights into business sectors.

EDA

There are 20 business categories. Services is one that has the highest number of businesses.

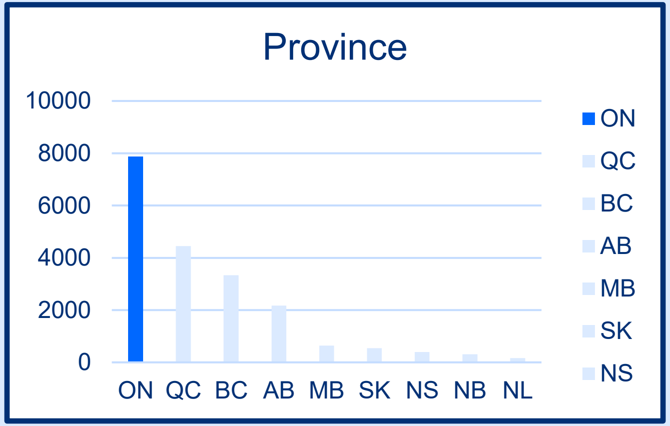
Categories	Response	Non Response
Healthcare	205	773
Manufacturing	299	1062
Religious	163	900
Retail	342	1513
Services	3236	13781
Grand Total	4245	18029

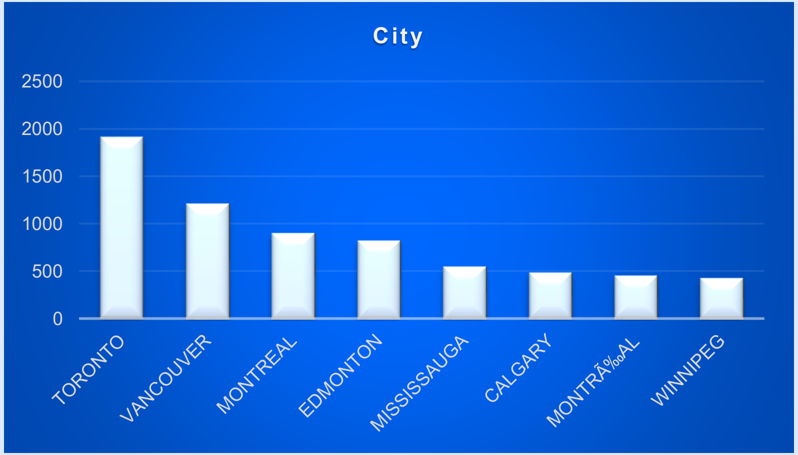




EDA

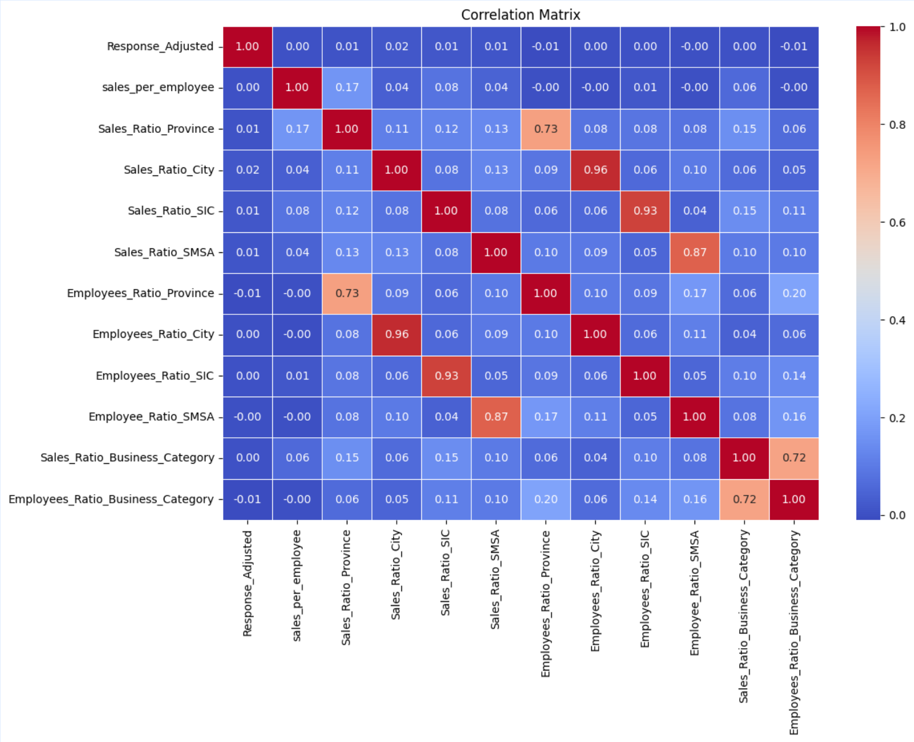
Province	Response.
ON	7876
QC	4448
BC	3339
AB	2171
MB	642
SK	537
NS	390
NB	305
NL	167
Grand Total	**19875**



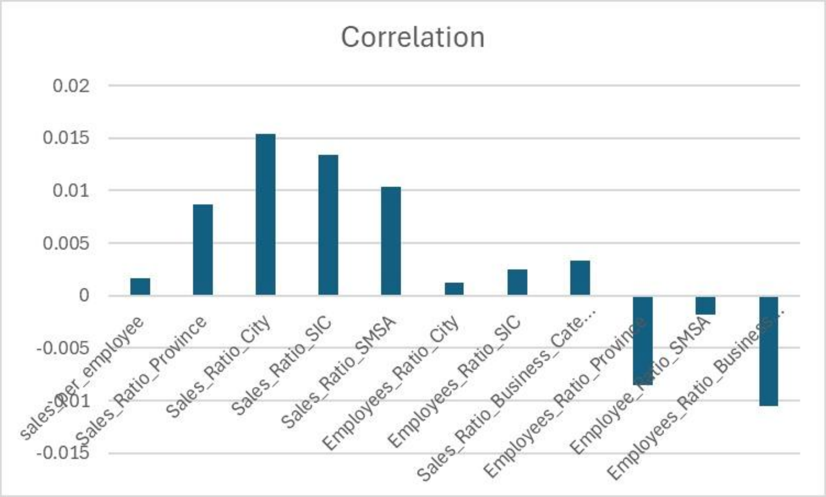


Row Labels	Response_
CALGARY	484
EDMONTON	824
MISSISSAUGA	551
MONTREAL	900
MONTRAL	457
TORONTO	1912
VANCOUVER	1214
WINNIPEG	426
Grand Total	**6768**

EDA



	Response_Adjusted	sales_per_employee	Sales_Ratio_Province	Sales_Ratio_City	Sales_Ratio_SIC	Sales_Ratio_SMSA	Employees_Ratio_Province	Employees_Ratio_City	Employees_Ratio_SIC	Employee_Ratio_SMSA	Sales_Ratio_Business_Category	Employees_Ratio_Business_Category
Response_Adjusted	1.00	0.00	0.01	0.02	0.01	0.01	-0.01	0.00	0.00	-0.00	0.00	-0.01
sales_per_employee	0.00	1.00	0.17	0.04	0.08	0.04	-0.00	-0.00	0.01	-0.00	0.06	-0.00
Sales_Ratio_Province	0.01	0.17	1.00	0.11	0.12	0.13	0.73	0.08	0.08	0.08	0.15	0.06
Sales_Ratio_City	0.02	0.04	0.11	1.00	0.08	0.13	0.09	0.96	0.06	0.10	0.06	0.05
Sales_Ratio_SIC	0.01	0.08	0.12	0.08	1.00	0.08	0.06	0.06	0.93	0.04	0.15	0.11
Sales_Ratio_SMSA	0.01	0.04	0.13	0.13	0.08	1.00	0.10	0.09	0.05	0.87	0.10	0.10
Employees_Ratio_Province	-0.01	-0.00	0.73	0.09	0.06	0.10	1.00	0.10	0.09	0.17	0.06	0.20
Employees_Ratio_City	0.00	-0.00	0.08	0.96	0.06	0.09	0.10	1.00	0.06	0.11	0.04	0.06
Employees_Ratio_SIC	0.00	0.01	0.08	0.06	0.93	0.05	0.09	0.06	1.00	0.05	0.10	0.14
Employee_Ratio_SMSA	-0.00	-0.00	0.08	0.10	0.04	0.87	0.17	0.11	0.05	1.00	0.08	0.16
Sales_Ratio_Business_Category	0.00	0.06	0.15	0.06	0.15	0.10	0.06	0.04	0.10	0.08	1.00	0.72
Employees_Ratio_Business_Category	-0.01	-0.00	0.06	0.05	0.11	0.10	0.20	0.06	0.14	0.16	0.72	1.00



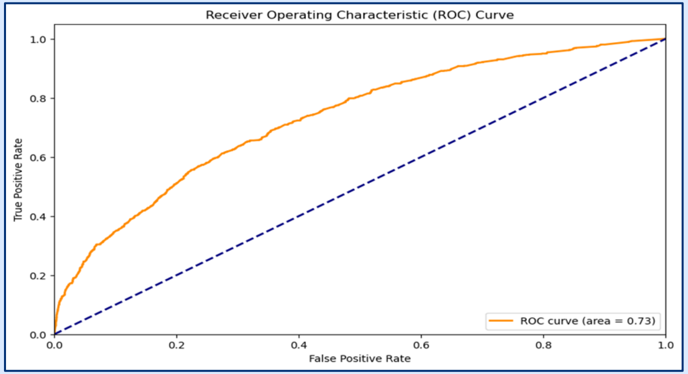
MODELING

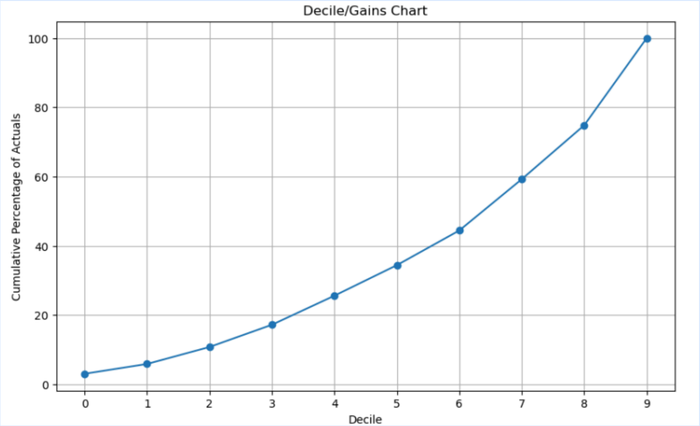
Model Type: Random Forest

Rationale:

- Handles mixed data types
- Reduces overfitting
- Robustness to Overfitting

MODEL PERFORMACE






```
Confusion Matrix:
[[2817  267]
 [ 619  299]]
```

77.83% Accuracy

52.65% Precision

62.24% AUC-ROC Score

Top Importance Influence:

Sales, Response_Ratio_SIC, and Response_Ratio_City

Moderate Importance Influence:

Features like sales_per_employee, Employees_Ratio_SIC, Sales_Ratio_SIC, Employees_Ratio_City, and Sales_Ratio_City

Lower Importance Influence:

City_Geo_Code, Response_Ratio_SMSA, Sales_Ratio_SMSA, Employee_Ratio_SMSA, Response_Ratio_Business_Category, Employees_Ratio_Business_Category, and Sales_Ratio_Business_Category

Least Importance Influence:

Response_Ratio_Province, Sales_Ratio_Province, and Employees_Ratio_Province

Feature	Importance
Sales	15%
Response_Ratio_SIC	14%
Response_Ratio_City	14%
sales_per_employee	11%
Employees_Ratio_SIC	7%
Sales_Ratio_SIC	7%
Employees_Ratio_City	5%
Sales_Ratio_City	5%
City_Geo_Code	5%
Response_Ratio_SMSA	3%
Sales_Ratio_SMSA	2%
Employee_Ratio_SMSA	2%
Response_Ratio_Business_Category	2%
Employees_Ratio_Business_Category	2%
Sales_Ratio_Business_Category	2%
Response_Ratio_Province	2%
Sales_Ratio_Province	1%
Employees_Ratio_Province	1%

LIMITATIONS

- Data Irregularities
- Missing Values
- Categorical and Ordinal Variables
- Scaling Issues

RECOMENDATIONS

- Targeted Marketing Campaign
- Tailored Messaging
- Budget Optimization
- Continuous Learning

THANK YOU

APPENDIX

Derived Variables:

•Sales_per_employee -
•Sales_Ratio_Province
•Sales_Ratio_City
•Sales_Ratio_SIC
•Sales_Ratio_SMSA
•Employees_Ratio_Province
•Employees_Ratio_City
•Employees_Ratio_SIC
•Employee_Ratio_SMSA
•Sales_Ratio_Business_Category
•Employees_Ratio_Business_Category

```python
# Select relevant features and target variable
X = df[['Sales', 'City_Geo_Code', 'sales_per_employee', 'Sales_Ratio_Province', 'Sales_Ratio_City',
        'Sales_Ratio_SIC', 'Sales_Ratio_SMSA', 'Employees_Ratio_Province', 'Employees_Ratio_City',
        'Employees_Ratio_SIC', 'Employee_Ratio_SMSA', 'Sales_Ratio_Business_Category',
        'Employees_Ratio_Business_Category', 'Response_Ratio_Province', 'Response_Ratio_City',
        'Response_Ratio_SIC', 'Response_Ratio_SMSA', 'Response_Ratio_Business_Category']]

y = df['Response_Adjusted']

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Replace infinite values with NaN
X_train.replace([np.inf, -np.inf], np.nan, inplace=True)

# Drop rows containing NaN values
X_train.dropna(inplace=True)

# Drop corresponding rows from y_train
y_train = y_train[X_train.index]

# Train a Random Forest classifier
model = RandomForestClassifier()
model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

# Evaluate the model's performance
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.7786106946526736
```